

ANTWERP INTL, Belgium

WMO: 064500

Lat: 51.1903N

Lon: 4.4578E

Elev: 39

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.0	24.6	14.7	11.6	23.3	18.9	14.3	26.5	27.2	48.3	24.6	49.3	4.9	70	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	85.6	69.0	81.6	67.2	78.1	65.4	70.5	82.2	68.6	79.0	66.8	75.9	6.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	97.3	75.8	64.7	91.6	73.7	63.1	86.5	71.7	34.4	82.4	32.8	79.4	31.4	76.1	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
21.2	18.6	16.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			15.8	92.4	5.0	3.0	12.2	94.6	9.3	96.4	6.5	98.1	2.9	100.3
			14.9	74.3	4.8	1.9	11.4	75.6	8.6	76.7	6.0	77.7	2.5	79.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	39.1	40.3	44.8	50.6	57.3	62.5	66.3	65.6	60.0	53.2	45.5	40.2	(6)
(7)		DBStd	11.50	7.71	7.02	6.06	6.22	6.18	5.50	5.44	4.93	4.70	5.87	6.11	7.45	(7)
(8)		HDD50	1372	341	276	178	67	8	0	0	0	0	35	157	310	(8)
(9)		HDD65	4897	803	692	625	432	252	113	49	50	162	367	584	768	(9)
(10)		CDD50	2172	3	4	18	85	233	375	504	485	301	134	23	7	(10)
(11)		CDD65	222	0	0	0	1	12	38	88	70	13	0	0	0	(11)
(12)		CDH74	1843	0	0	0	22	138	336	702	554	86	5	0	0	(12)
(13)		CDH80	504	0	0	0	1	22	86	211	171	13	0	0	0	(13)
(14)	Wind	WSAvg	7.6	9.0	8.7	8.6	7.3	7.3	7.2	6.9	6.5	6.6	7.3	7.6	8.6	(14)
(15)	Precipitation	PrecAvg	32.2	2.7	2.4	2.0	1.7	2.3	3.0	3.1	3.3	2.6	2.6	3.1	3.3	(15)
(16)		PrecMax	39.4	5.6	5.0	4.3	3.7	4.3	5.2	5.9	9.1	7.6	5.8	5.9	7.5	(16)
(17)		PrecMin	24.8	0.2	0.6	0.2	0.0	0.6	1.0	1.0	0.4	0.4	0.8	0.3	1.0	(17)
(18)		PrecStd	3.9	1.3	1.1	1.1	1.0	1.0	1.2	1.4	2.2	1.8	1.0	1.2	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.0	59.2	66.3	77.3	83.0	88.1	91.2	90.4	81.7	73.7	63.0	57.3	(19)
(20)		MCWB	52.1	50.1	54.1	60.9	65.9	69.9	70.8	71.3	67.7	63.3	57.3	53.2	(20)	
(21)		2%	DB	53.7	54.9	61.4	71.2	77.8	82.0	85.6	84.7	76.1	67.9	59.1	55.1	(21)
(22)		MCWB	50.4	49.7	52.5	58.0	62.5	67.5	69.2	68.8	65.1	60.7	54.9	51.8	(22)	
(23)		5%	DB	51.4	52.3	57.3	66.3	73.5	77.6	81.4	80.2	72.0	64.5	56.7	53.1	(23)
(24)		MCWB	48.3	48.3	50.5	55.6	60.9	65.2	67.2	67.1	62.7	58.7	53.1	50.0	(24)	
(25)		10%	DB	49.1	50.1	54.4	61.9	69.2	73.6	77.5	75.8	68.7	61.9	54.5	50.4	(25)
(26)		MCWB	46.2	46.7	49.2	53.2	59.0	63.0	65.6	65.0	60.9	57.2	51.3	47.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	52.5	56.2	62.9	67.8	72.0	72.9	73.2	69.2	63.9	58.0	54.2	(27)
(28)		MCDWB	55.0	56.3	62.9	73.7	79.4	84.6	87.1	85.6	78.4	71.1	61.1	56.2	(28)	
(29)		2%	WB	50.8	50.4	53.7	58.7	64.7	68.7	70.4	70.2	66.3	61.4	55.6	52.3	(29)
(30)		MCDWB	53.2	53.6	59.2	69.0	73.9	79.1	82.6	81.3	74.3	66.8	58.5	54.8	(30)	
(31)		5%	WB	48.5	48.9	51.6	56.2	61.9	66.4	68.6	68.1	63.9	59.6	53.7	50.2	(31)
(32)		MCDWB	51.1	52.0	56.4	64.7	70.7	75.9	79.4	77.8	69.8	63.8	56.4	52.8	(32)	
(33)		10%	WB	46.4	47.0	49.7	54.0	59.7	64.1	66.7	66.0	62.1	57.9	51.4	47.9	(33)
(34)		MCDWB	48.9	50.0	53.6	61.1	68.0	71.8	75.5	73.9	67.4	61.3	54.1	50.4	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.7	10.3	13.1	16.2	16.1	15.9	15.7	16.0	14.7	12.9	9.8	8.2	(35)	
(36)		5% DB	MCDBR	9.8	13.5	19.3	24.6	24.1	23.4	23.3	24.4	20.5	16.8	11.7	9.2	(36)
(37)		MCWBR	8.5	9.8	12.2	13.6	11.7	11.8	10.2	11.1	11.0	10.4	9.0	7.9	(37)	
(38)		5% WB	MCDBR	9.3	11.2	16.7	21.6	20.7	21.4	20.9	21.8	18.4	15.4	10.9	8.7	(38)
(39)		MCWBR	8.6	8.9	11.2	12.6	11.4	11.8	10.1	11.0	10.6	10.1	9.0	7.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.356	0.390	0.423	0.433	0.429	0.434	0.424	0.397	0.370	0.346	0.331	(40)	
(41)		taud	2.369	2.336	2.242	2.180	2.199	2.231	2.220	2.267	2.331	2.393	2.377	2.340	(41)	
(42)		Ebn at Noon	214	241	253	258	261	263	259	255	249	233	206	193	(42)	
(43)		Edh at Noon	20	27	36	42	43	42	42	38	32	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	241	456	825	1321	1580	1713	1613	1376	1023	601	283	186	(44)	
(45)		RadStd	30	89	117	145	139	143	192	129	97	49	30	25	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page